

Yichao Jin, Ph.D.

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Computational Social Science / Behavioral AI / Structural Causal Modeling / Causal Inference

Education

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| University of Texas at Dallas | May 2026 |
| Ph.D. in Public Policy and Political Economy | |
| <ul style="list-style-type: none">- Dissertation: “Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan” — ProQuest- Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong.- Dean of Graduate Education Dissertation Research Award (2025). | |
| University of Texas at Dallas | May 2026 |
| M.S. in Social Data Analytics and Research | |
| University of Queensland | 2019 |
| Master of Development Economics | |
| University of California, Riverside | 2017 |
| B.A. in Economics | |

Research Agenda

My research develops causally-constrained AI agents for behavioral simulation and policy analysis. I integrate discrete choice experiments, structural causal modeling, and large language models to build Behavioral Digital Twins—empirically-grounded synthetic agents that generate behavior within estimated human choice frontiers rather than from LLM training priors. This framework enables pre-screening of behavioral interventions and policy designs before field deployment, with applications to health behavior, education, and social policy.

Working Papers

- “Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework”**
Working Paper (Target: J. Biomedical Informatics) — [SSRN Link](#). Introduces a DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule, the Static-BDT Anchor reduces waiting-time monotonicity violations from 62.5% to 4.2% and improves Spearman rank correlation from -0.024 to 0.952 . Presented at DAISY 2026 (ACM CHIL 2026) and AIXPH 2026.
- “Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects”**
Working Paper — [SSRN Link](#). Audit framework measuring value sensitivity in LLM-generated policy simulations. Tests nine LLMs across six model families through policy-value orientation batteries, narrative frame-decomposition, and within-model frame-manipulation experiments (N=2,160 observations).

3. **“Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust”**
Under Review at Economic Analysis and Policy — [Link to Full Draft](#). Structural DCE framework measuring interaction between temporal barriers and institutional trust in vaccination decisions.
4. **“The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount”**
Manuscript in Preparation (Expected submission: August 2026).

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Conference Presentations

- **AIxPH 2026 (1st Annual Conference on AI and Public Health, Baltimore, MD)**: Empirical-Frontier Regularization for LLM Synthetic Agents — Poster Presentation.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.

AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.

Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.

Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
 - **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
 - **2016** Omicron Delta Epsilon, International Honor Society for Economics
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REFERENCES

Dohyeong Kim (Chair)

University of Texas at Dallas
Senior Associate Dean of Graduate Education
Robert E. Holmes Jr. Professor of Public Policy,
GIS & Social Data Analytics
Team Lead, SHARP
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